Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[progressbar=frametitle]beamerthememetropolis
[]beamercolorthemewolf appendixnumberbeamer
booktabs [scale=2]ccicons
pgfplots dateplot
xspace

Subtitle

Name

19 Pebruari 2025

North Carolina State University

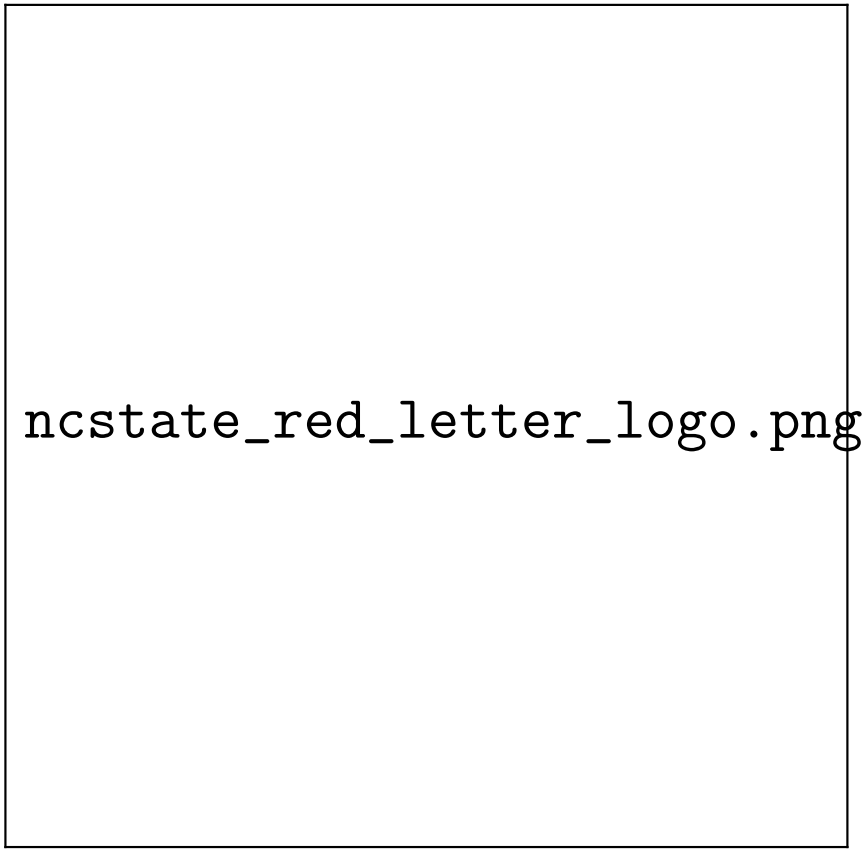


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Metropolis

The **metropolis**theme is a Beamer theme with minimal visual noise inspired by the hsrm Beamer Theme by Benjamin Weiss.
Enable the theme by loading

```
\documentclass{beamer}
\usetheme{metropolis}
```

Note, that you have to have Mozilla's *Fira Sans* font and XeTeX installed to enjoy this wonderful typography.

Sections

Sections group slides of the same topic

```
\section{Elements}
```

for which **metropolis**provides a nice progress indicator . . .

Metropolis titleformats

metropolissupports 4 different titleformats:

- Regular
- Smallcaps
- allsmallcaps
- ALLCAPS

They can either be set at once for every title type or individually.

titleformat frame=smallcaps

Small caps

This frame uses the `smallcaps` titleformat.

Potential Problems

Be aware, that not every font supports small caps. If for example you typeset your presentation with pdfTeX and the Computer Modern Sans Serif font, every text in smallcaps will be typeset with the Computer Modern Serif font instead.

titleformat frame=allsmallcaps

All small caps

This frame uses the `allsmallcaps` titleformat.

Potential problems

As this titleformat also uses smallcaps you face the same problems as with the `smallcaps` titleformat. Additionally this format can cause some other problems. Please refer to the documentation if you consider using it.
As a rule of thumb: Just use it for plaintext-only titles.

titleformat frame=allcaps

All caps

This frame uses the `allcaps` titleformat.

Potential Problems

This titleformat is not as problematic as the `allsmallcaps` format, but basically suffers from the same deficiencies. So please have a look at the documentation if you want to use it.

Typography

```
The theme provides sensible defaults to
\emph{emphasize} text, \alert{accent} parts
or show \textbf{bold} results.
```

becomes

The theme provides sensible defaults to *emphasize* text, accent parts or show **bold** results.

Font feature test

- Regular
- *Italic*
- SmallCaps
- **Bold**
- ***Bold Italic***
- **Bold SmallCaps**
- `Monospace`
- *`Monospace Italic`*
- `Monospace Bold`
- *`Monospace Bold Italic`*

Lists

Items

- Milk
- Eggs
- Potatos

Enumerations

1. First,
2. Second and
3. Last.

Descriptions

PowerPoint Meeh.

Beamer Yeeeha.

Animation

- This is important

Animation

- This is important
- Now this

Animation

- This is important
- Now this
- And now this

Animation

- This is really important
- Now this
- And now this

Figures

coordinate](0,0)coordinate(A)++(90:
5cm)coordinate(B)++(0:5cm)coordinate(C)++(-90:
5cm)coordinate(D);\drawfill=alerted text.fg!20] (A) – (B) – (C)
–(D) – cycle; in 1,...,40
coordinate]coordinate(X)at(A){};\pathcoordinate] (A) – (B)
coordinate[pos=.10](A) – (C) coordinate[pos=.10](B) – (D)
coordinate[pos=.10](C) – (X) coordinate[pos=.10](D); [fill=alerted
text.fg!40] (A)–(B)–(C)– (D) – cycle;

Gambar: Rotated square from texample.net.

Tables

Tabel: Largest cities in the world (source: Wikipedia)

City	Population
Mexico City	20,116,842
Shanghai	19,210,000
Peking	15,796,450
Istanbul	14,160,467

Blocks

Three different block environments are pre-defined and may be styled with an optional background color.

Default
Block content.

Alert
Block content.

Example
Block content.

block=fill

Default
Block content.

Alert
Block content.

Example
Block content.

Math

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

Line plots

[mlineplot, width=0.9height=6cm,]
sin(deg(x)); +[samples=100] sin(deg(2*x));

Bar charts

[mbarplot, xlabel=Foo, ylabel=Bar, width=0.9height=6cm,]
plot coordinates (1, 20) (2, 25) (3, 22.4) (4, 12.4); plot coordinates (1, 18) (2, 24) (3, 23.5) (4, 13.2); plot coordinates (1, 10) (2, 19) (3, 25) (4, 15.2);
lorem, ipsum, dolor

Quotes

Veni, Vidi, Vici

Frame footer

metropolisdefines a custom beamer template to add a text to the footer. It can be set via

```
\setbeamertemplate{frame footer}{My custom footer}
```

References

Some references to showcase [allowframebreaks] [?, ?, ?, ?, ?]

Summary

Get the source of this theme and the demo presentation from

`github.com/matze/mtheme`

The theme *itself* is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

Questions?

Backup slides

Sometimes, it is useful to add slides at the end of your presentation to refer to during audience questions.
The best way to do this is to include the `appendixnumberbeamer` package in your preamble and call `\appendix` before your backup slides.
metropoliswill automatically turn off slide numbering and progress bars for slides in the appendix.

References I